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Cooling performance indicator

Abstract

A cooling system for a vehicle includes a processing device configured to determine a coolant system status based on a coolant temperature and determine whether the coolant system status exceeds a coolant system status threshold. The processing device is also configured to determine a thermal load status of at least one electrical component of the vehicle and determine whether the thermal load status exceeds a thermal load status threshold. The processing device is also configured to output a first cooling performance indicator in response, at least in part, to determining that the coolant system status exceeds the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold, the first cooling performance indicator indicating an issue with the cooling system.

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Background/Summary

FIELD

(1) This disclosure relates to cooling systems for electrical components in vehicles, and more particularly to cooling performance indicators for cooling systems in electric vehicles and hybrid vehicles.

BACKGROUND

(2) Electric vehicles (EVs) and hybrid vehicles (HVs), such as plug-in hybrid vehicles (PHEVs) and mild hybrid electric vehicles (MHEVs), are increasingly commonplace. Such vehicles include multiple electric components, such as electric motors, starter-generators, inverters, and controllers, that generate large amounts of heat and require cooling to remain functional and efficient. Typically, as the temperatures of such components increase beyond certain temperatures, the components become derated such that their functional efficiency is reduced, and the components can ultimately temporarily or permanently fail due to excessive heat. This reduces the efficiency of EVs, and can result in higher reliance on the combustion engine in HVs. Such negative outcomes serve to undercut the primary purpose of EVs and HVs, which is to improve energy efficiency.

(3) Various regulatory boards, such as the California Air Resource Board (CARB), have

implemented a requirement for performing a functional check of the cooling system performance and outputting a notification regarding such performance. While existing systems comply with such regulations regarding notification of cooling system performance, there is a desire to provide information on the performance of the cooling system with improved detail or specificity.

SUMMARY

(4) To address these and other concerns, an improved cooling system for electrical components in a vehicle is disclosed. In accordance with various embodiments, cooling system includes at least processing device that is configured to determine a coolant system status based, at least in part, on a coolant temperature of coolant in the cooling system, and determine whether the coolant system status exceeds a coolant system status threshold. The processing device is also configured to determine a thermal load status of at least one electrical component of the vehicle cooled by the cooling system and determine whether the thermal load status exceeds a thermal load status threshold. The electrical component may be an inverter, a controller, a charge control unit, an autonomous driving control unit, a battery, or a starter-generator. The processing device is also configured to output a first cooling performance indicator in response, at least in part, to determining that the coolant system status exceeds the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold, the first cooling performance indicator indicating an issue with the cooling system.

(5) In some embodiments, the cooling system also includes a coolant temperature sensor that measures the coolant temperature of the coolant in the cooling system at an exit of a coolant cooler. The processing device is then further configured to receive the coolant temperature from the coolant temperature sensor and determine the coolant system status based, at least in part, on the coolant temperature and also, in some embodiments, an ambient temperature and a vehicle speed. In some embodiments, the cooling system also includes an electrical component temperature sensor that measures a temperature of the electrical component. The processing device is then also configured to receive the temperature of the electrical component and determine the thermal load status based on the temperature of the electrical component.

(6) In some embodiments, the cooling system also includes a cooling system component, such as a cooling fan, a coolant pump, or a shutter flap. The processing device is further configured to determine that the cooling system component has failed, output a second cooling performance indicator in response, at least in part, to determining that the cooling system component has failed, the second cooling performance indicator indicating a cooling system component failure, and prevent output of the first cooling performance indicator. In this manner, the system may provide a more specific performance indicator that provides information that one of the cooling components of the cooling system has failed, rather than the general first cooling performance indicator for the entire cooling system.

(7) In certain embodiments, the processing device may be configured to output a third cooling performance indicator in response, at least in part, to determining that the coolant system status does not exceed the coolant system status threshold and that the thermal load status of the at least one electrical component exceeds the thermal load status threshold. The third cooling performance indicator indicates an issue with cooling of the at least one electrical component. In this manner, individual third cooling performance indicators specific to each electrical component may be output that indicate an issue with cooling of that electrical component rather than the general first cooling performance indicator for the entire cooling system. In various approaches, the processing device is configured to output the first, second, and/or third cooling performance indicators by setting a diagnostic trouble code in an on-board diagnostics (OBD) system of the vehicle.

(8) As such, an improved cooling system provides additional details regarding issues within the cooling system, including specific indications of particular component cooling failures. This leads to greater efficiency in diagnosing and repairing such cooling systems, as well as improved detection of coolant system issues.

(9) Other objects, advantages and novel features of the present invention will become apparent from the following detailed description of one or more preferred embodiments when considered in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic representation of a cooling system in accordance with various embodiments;

(2) FIG. **2** is another schematic representation of another cooling system in accordance with various embodiments;

(3) FIG. **3** is another schematic representation of another cooling system in accordance with various embodiments;

(4) FIG. **4** is logic flow diagram illustrating various functions and methods of outputting a coolant performance indicator in accordance with various embodiments;

(5) FIG. **5** is another logic flow diagram illustrating variations of the functions and methods disclosed in FIG. **4** in accordance with various embodiments; and

(6) FIG. **6** is another logic flow diagram illustrating variations of the functions and methods disclosed in FIG. **5** in accordance with various embodiments.

DETAILED DESCRIPTION OF THE DRAWINGS

(7) Existing cooling systems utilize information from one electrical component, such as the inverter, to determine cooling system performance. For example, existing systems may rely principally on the temperature of the inverter, analyzing temperature increases and decreases of the inverter. However, typically, cooling systems have parallel cooling paths, and such analysis based principally on the temperature from one component (e.g., the inverter) in one of those parallel cooling paths may provide only limited information on the performance of the cooling system as a whole. For example, if cooling path that includes the inverter is blocked, then existing systems could identify that there is an issue with the cooling system as a whole (e.g., based on the increased temperature of the inverter), but would only indicate that the entire cooling system is experiencing problems and may not be able to indicate specifically that the cooling path of the inverter is having an issue. Similarly, if a different component (e.g., in a different branch of the cooling system) is experiencing cooling problems, while the inverter does not experience such cooling problems, the existing system may not be able to identify that such an issue with the cooling system exists as the inverter is operating without issue. Similarly, because the cooling system performance is based principally on the temperature of this one component (e.g., the inverter), it is also difficult to determine if a temperature increase of that component is due to a cooling system failure or due to the operational state or function of that component.

(8) FIG. **1** is a schematic representation of a cooling system **100** implementing improved cooling performance analysis and reporting in accordance with various embodiments. In various embodiments, the cooling system **100** of FIG. **1** may be for an electric vehicle (EV) or a hybrid vehicle (HV), such as a plug-in hybrid electric vehicle (PHEV). The cooling system **100** includes a coolant circuit **102** that includes a plurality of pathways (e.g., hoses, etc.) through which coolant fluid can pass, thereby cooling a plurality of electrical components of a vehicle. These pathways may include multiple parallel branches through which the coolant can pass. The cooling system **100** may include a coolant cooler **104** through which heated coolant within the circuit **102** passes and is cooled, for example, by ambient air. A fan **106** may be included, which moves cool air across the coolant cooler **104** to further reduce the temperature of the coolant therein. An air vent shutter **108** (including a shutter controller) may also be included, which can be opened or closed to allow more or less air to pass through the coolant cooler **104**. A compensation tank **110** may also be

included within the system **100**. A coolant pump **112** pumps the coolant through the circuit **102** so that the cooled coolant passes through the pathways and hoses, past the various electrical component of the vehicle such that the coolant absorbs the heat generated by those electrical components, and back to the coolant cooler **104** to remove the heat from the coolant and cool the coolant.

(9) In various embodiments, the cooling system **100** may include an intercooler **114**, which may include a valve **116** to control the flow of coolant to the intercooler. The cooling system **100** may also include an inverter **118**, a charge control unit (CCU) **120**, an autonomous driving controller **122**, and a water cooler condenser **124**. The inverter **118** may convert DC electric power (e.g., stored in a battery) into AC power via semiconductor switches to control and power an electric motor of the vehicle. The CCU **120** may control charging of the inverter **118** or other components, and may also control cooling of the entire electrical cooling system **100**. The autonomous driving controller **122** may control and implement all or some autonomous driving functions of the vehicle. Each of these components generate heat which must be removed in order for the component to remain efficient and functioning. The coolant in the circuit **102** may pass through or over each of these electrical components to remove the heat generated by these electrical components.

(10) The cooling system **100** may include or be coupled to an engine control unit (ECU) **126**. The ECU **126** may include a (e.g., one or more) processing device **128** that is coupled to a (e.g., one or more) memory **130**, for example, with a data bus. The processing device **128** may be a Central Processing Unit (CPU), microcontroller, or a microprocessor, and/or may include or be implemented with an Application Specific Integrated Circuit (ASIC), Programmable Logic Device (PLD), or Field Programmable Gate Array (FPGA). The processing device **128** and/or the ECU **126** as a whole may also be implemented with circuitry that includes discrete logic or other circuit components, including analog circuit components, digital circuit components or both. The circuitry may include discrete interconnected hardware components or may be combined on a single integrated circuit die, distributed among multiple integrated circuit dies, or implemented in a Multiple Chip Module (MCM) of multiple integrated circuit dies in a common package, as examples. The memory **130** may comprise a flash memory, a Random Access Memory (RAM), a Read Only Memory (ROM), an Erasable Programmable Read Only Memory (EPROM), a Hard Disk Drive (HDD), other magnetic or optical disk, or another machine-readable nonvolatile medium or other tangible storage mediums other than a transitory signal. The memory **130** may store therein software modules, code, and/or instructions that, when executed by the processing device **128**, cause the processing device **128** to implement some or all of the processes described herein or illustrated in the drawings. The memory **130** may also store other data for use by the processing device **128** and/or historical information regarding the operation of the ECU **126**, other components, and or the vehicle as a whole. The CCU **120**, autonomous driving controller **122**, power control unit **204** (see FIGS. **2** and **3**), and starter-generator **302** (see FIG. **3**) may each also include a similar processor and memory as discussed directly above.

(11) In various embodiments, the ECU **126** provides an On-Board Diagnostics (OBD) function, which can output diagnostic trouble codes (DTC) to a compatible reader system. These DTC codes may be stored in the memory **130**, for example as flags, upon detection of a triggering condition and may be saved until cleared by a compatible reader system or expiration of time. In various embodiments herein, a cooling performance indicator is provided as a DTC code through the OBD system. In various approaches, the cooling performance indicator may indicate an issue with the cooling system as a whole, with a cooling system component (such as the fan **106**, shutter **108**, or pump **112**), which may include, for example, different DTC codes for each identified issue type. These different cooling performance indicators may be triggered according to different cooling system performance analysis processes as disclosed herein.

(12) As part of the processes of analyzing the performance of the cooling system **100**, additional data is captured, determined, transferred between the different electrical components (e.g., via a

vehicle bus system such as a CAN bus or via other known data transmission methods), and processed. For example, the cooling system **100** also includes one or more coolant temperature sensors **132** within the circuit **102**. In various embodiments, the coolant temperature sensor **132** is located at or near the outlet of the coolant cooler **104** such that it measures the temperature of the coolant as it exits the coolant cooler **104** or shortly thereafter. However, the coolant temperature sensor **132** may be located at other or additional locations within the circuit **102**. Similarly, the inverter **118** may include one or more inverter temperature sensors **134** that measure the temperature of the inverter **118**. Other electrical components may include similar temperature sensors to measure the temperature of each component, including, for example, the CCU **120**, the autonomous driving controller **122**, the power control unit **204** (see FIGS. **2** and **3**), and/or the battery **202** (which includes battery temperature sensor **206**) (see FIGS. **2** and **3**).
(13) As shown in FIG. **1**, the coolant temperature sensor **132** may provide a temperature of the coolant (or other related information) to the ECU **126**, though it may provide such data to other electrical components such as the CCU **120**. Similarly, the inverter **118** may provide an inverter temperature (taken with the inverter temperature sensor **134**) and/or a thermal load status of the inverter **118** derived from the inverter temperature to the ECU **126** and/or the CCU **120**. Similarly, the autonomous driving controller **122**, the power control unit **204** (see FIGS. **2** and **3**), and/or the battery **202** (see FIGS. **2** and **3**) may also provide temperatures and/or determined thermal load statuses for each component to the CCU **120** and/or the ECU **126**. Similarly still, the CCU **120** may provide its temperature and/or determined thermal load status to the ECU **126**. In certain embodiments, all of some of the electrical components provide temperatures and/or determined thermal load statuses for each component to the CCU **120**, and the CCU **120** in turn provides that data or other data related to the temperatures and/or determined thermal load statuses, to the ECU **126**.
(14) Additionally, cooling system components, such as the fan **106**, the controller of the air vent shutter **108**, the coolant pump **112**, or the valve **116** may provide operational status information to the CCU **120** and/or the ECU **126**. Specifically, the fan **106**, the controller of the air vent shutter **108**, the coolant pump **112**, and/or the valve **116** may provide present functional status info (e.g., present speed of the fan **106**, opening amount of the shutter **108**, speed of the coolant pump **112**, open amount of the valve **116**), as well as diagnostic fault information indicating whether the fan **106**, the air vent shutter **108**, the coolant pump **112**, or valve **116** are operating correctly or are failing or have failed.
(15) FIG. **2** is another schematic representation of a cooling system **200** that is a variation of the cooling system **100** of FIG. **1** in accordance with various embodiments. FIG. **3** is another schematic representation of another cooling system **300** that is a variation of the cooling system **100** of FIG. **1** and the cooling system **200** of FIG. **2** in accordance with various embodiments. The cooling systems **200** and **300** of FIGS. **2** and **3** may be for mild hybrid electric vehicle (MHEV), and may include a lower voltage battery **202** (e.g., 48V) to provide power. A power control unit **204** is also included, which controls the flow of power to and from the battery **202**. In various embodiments, the battery **202** and the power control unit **204** may be cooled by the cooling system, as is shown by the coolant pathways of the cooling circuit **102**. As mentioned above, the battery **202** and the power control unit **204** may include temperature sensors, such as battery temperature sensor **206**, to measure the temperature of each component. Similarly, the battery **202** and the power control unit **204** may provide temperatures and/or determined thermal load statuses for each component to the ECU **126**.
(16) While the cooling system **200** of FIG. **2** includes the inverter **118**, the cooling system **300** of FIG. **3** may include a starter-generator **302** that receives power from and generates power to be stored in the battery **202**. In various embodiments, the starter-generator **302** may provide a torque signal to the ECU **126** (or to other controllers) indicating the amount of torque the starter-generator **302** is outputting. Similarly, the starter-generator **302** may also provide a rotational speed signal

and/or a temperature signal to the ECU **126** (or to other controllers). The remainder of the cooling systems **200** and **300** of FIGS. **2** and **3** are similar to the system **100** of FIG. **1**, and similar or identical components are numbered the same in each figure.

(17) Operational processes and configurations of the processing devices (e.g., the processing device **128**) of the cooling system **100** are provided below. Specifically, processes for triggering output of a cooling performance indicator (e.g., a DTC in certain embodiments) are disclosed. Although cooling system **100** is reference below, it is understood that some or all of these processes are implemented in each cooling system **100**, **200**, and **300**. Also, although the processor **128** of the ECU **126** is generally described as being configured to execute the processes discussed herein, other processing devices of the various electrical components described herein, or other processing devices of other control units or electrical systems, may likewise be configured to execute all or some of the disclosed processes. Similarly, execution of the processes disclosed herein may involve different processing devices of different electrical components executing different portions of the disclosed processes.

(18) FIG. **4** is logic flow diagram illustrating various functions and methods of outputting a coolant performance indicator, which the one or more processing devices of the system may be configured to execute, in accordance with various embodiments. As shown at **404**, the at least one processing device (e.g., processing device **128**) may be configured to determine a coolant system status based, at least in part, on a coolant temperature of coolant in the cooling system **100**. The at least one processing device may receive the coolant temperature from the coolant temperature sensor **132**, which may be configured to measure the coolant temperature of the coolant in the cooling system **100** at an exit of a coolant cooler **104**. The processing device may determine the coolant system status based, at least in part, on the coolant temperature, an ambient temperature, and a vehicle speed. For example, the cooling system status may be one or more levels (e.g., four levels, including cold, OK, full workload, and critical), which status level may be determined based on a mapping of the coolant temperature, an ambient temperature, and a vehicle speed. The mapping may be determined based on a lookup table stored in the memory (e.g., memory **130**) or may be continuously or periodically computed based on one or more equations. In a specific approach, coolant temperature thresholds are set to define the different status levels, and the thresholds are adjusted based on the vehicle speed and the ambient temperature. These thresholds may also be set based in part on the mileage of the vehicle.

(19) Next, at **406**, the at least one processing device is configured to determine a thermal load status of at least one electrical component of the vehicle cooled by the cooling system **100**, which may include an inverter **118**, a controller, a charge control unit **120**, an autonomous driving controller **122**, a battery **202**, a power control unit **204**, a starter-generator **302**, or another component. This may be performed by the processing device of the ECU **126** or a processing device of the CCU **120**. Alternatively, this may be performed by a processing device associated or included with the electrical component may be configured to determine the thermal load status of that electrical component itself. In this case, the individual electrical components may then communicate its determined thermal load status to the CCU **120** and/or the ECU **126** for further processing. The individual electrical components may include an electrical component temperature sensor configured to measure a temperature of the electrical component, such as, for example, the inverter temperature sensor **134** or the battery temperature sensor **206**. The at least one processing device is configured to receive the temperature of the at least one electrical component and determine the thermal load status based on the temperature of the at least one electrical component.

(20) Like the coolant system status, the thermal load status may have one or more status levels which are mapped to the measured temperature (or multiple measured temperatures of the component if more than one sensor is included), for example, cold, OK, at capacity, critical, and derating. Each electrical component may have its own mapping, particularly for a critical or derating status level. For example, the battery **202** may suffer derating at 55° C., while the inverter

118 may suffer derating at 70-90° C. As such, they have different individual temperature mappings for their thermal load status.

(21) During normal operation, these thermal load statuses may be used to control the cooling system **100**, including the speed of the coolant pump **112** or the fan **106**, or the opening or closing of the shutter **108**. For example, when one or more component thermal load statuses indicate that the component is approaching or at a critical level, then the cooling system **100** may maximize the speed of the coolant pump **112** and the fan **106**, and may ensure the shutter **108** is fully open. If such operations do not lower the thermal load status of the component, then the electrical component may then enter a state that impacts efficiency and the at least one processing device (e.g., the processing device **128** of the ECU **126**) may have to flag the issue using a cooling performance indicator (e.g., a DTC).

(22) Next, at **408**, the at least one processing device is configured to determine whether the coolant system status exceeds a coolant system status threshold. In a specific example, it determines if the coolant system status is at a critical level. The coolant system may reach a critical level in response to different issues within the coolant system, such as for example, a blocked cooler or a wrong cooling fluid, or other issues that affect operations of the cooling system. Also, at **410**, the at least one processing device is configured to determine whether a thermal load status for one of the electrical components cooled by the cooling system exceeds a thermal load status threshold. If the answer to both of these inquiries is yes (at **412**), then the at least one processing device may determine that a fault has occurred and will determine that the fault is pending. This pending fault can lead to the processing device outputting and/or storing in the memory a first cooling performance indicator (e.g., DTC **416**) indicating an issue with the cooling system **100**. Put another way, the processing device can output the first cooling performance indicator (e.g., DTC **416**) in response, at least in part, to determining that the coolant system status exceeds the coolant system status threshold (at **408**) and that the thermal load status exceeds the thermal load status threshold (at **410**). In various approaches, for the first cooling performance indicator (e.g., DTC **416**) to be output, the processing device must determine that a fault pending duration of time exceeds a fault pending threshold time. In this instance, then the processing device may then output the first cooling performance indicator (e.g., DTC **416**) in response, at least in part, to also determining that the fault pending duration of time exceeds the fault pending threshold time (at **414**).

(23) Thus, in an instance where the coolant system status is critical, and at least one component thermal load status is critical, and they both remain critical for a fault pending threshold time (e.g., 1 minute) to ensure that the problem is not merely a short peak, the processing device will output the first cooling performance indicator, for example, by setting a DTC in an OBD system of the vehicle.

(24) In certain embodiments, at **402**, the at least one processing device may also determine an operational status of a cooling system component (such as the fan **106**, shutter **108**, or pump **112**), specifically determining if a cooling system component has failed. If such a cooling system component has failed, the at least one processing device may output a second cooling performance indicator in response, at least in part, to determining that the cooling system component has failed. The second cooling performance indicator may be a DTC **420** indicating a cooling system component failure. In various embodiments, the second cooling performance indicator may one or more of a plurality of various DTCs that separately relate to the different components or issues so that pinpointing a problem is possible. Further, the DTC for each component may include information indicating what kind of error exists, for example, such as an electrical issue (short to ground/plus) or plausibility errors (offset/stuck/out of range). In various examples, each of these different failure types may have its own DTC. In one approach, the at least one processing device may also determine, at **418**, that the cooling system component has failed for a component failure duration of time that exceeds a component failure pending threshold time. In this approach, the at least one processing device is configured to then output the second cooling performance indicator

in response, at least in part, to also determining that the component failure duration of time exceeds the component failure threshold time. This again ensures that the fault detected with the cooling system component is not merely a transitory issue.

(25) Additionally, the processing device may be configured to prevent or block output of the first cooling performance indicator (e.g., DTC **416**). This is because the root cause of the coolant system issue may be limited to failure of that particular cooling system component, rather than a different issue with the cooling system, such as a blocked pipe. Thus, instead of the general first cooling performance indicator (e.g., DTC **416**) for the entire cooling system, a more specific second cooling performance indicator (e.g., DTC **420**) can be provided that indicates there is an issue with the cooling system component.

(26) FIG. **5** is another logic flow diagram illustrating additional functions and methods, which the one or more processing devices of the system may be configured to execute, in accordance with various embodiments. FIG. **5** merely adds features to those shown in FIG. **4**, and as such, FIG. **4** is wholly encompassed by FIG. **5**. In FIG. **5**, the added functions are to determine cooling system failures specific to each electrical component cooled by the cooling system. More specific third cooling performance indicators specific to each electrical component may be output that indicate an issue with cooling of that electrical component. Such component-specific cooling issues may result from, for example, a blocked pathway or leak on a particular coolant branch for that component, or other issues that affect the cooling of that electrical component, but may not generally affect the cooling of other electrical components.

(27) As shown at **408**, the at least one processing device (e.g., processing device **128**) may be configured to determine that the coolant system status does not exceed the coolant system status threshold. However, even in a situation where the coolant system status is not critical (e.g., the coolant temperature is not too high for the ambient temperature and speed of the vehicle), the at least one processing device may determine that one or more electrical components has a thermal load status that exceeds the threshold (e.g., critical). In this case, the at least one processing device may output a third cooling performance indicator (e.g., DTC **506**, **512**, **518**) in response, at least in part, to determining (at **502**, **508**, or **514**) that the coolant system status does not exceed the coolant system status threshold and that a thermal load status of the at least one electrical component does exceed the thermal load status threshold.

(28) Each electrical component may trigger its own third cooling performance indicator (e.g., DTC). For example, as shown in FIG. **5**, if the thermal load status of the inverter **118** exceeds the threshold for the inverter **118** (e.g., critical) at **410**, while the coolant system status remains under the critical threshold at **408**, then, at **502**, the at least one processing device will determine that a fault is pending with respect to the cooling of the inverter **118**. Similarly, if the thermal load status of the battery **202** exceeds the threshold for the battery **202** (e.g., critical) at **410**, while the coolant system status remains under the critical threshold at **408**, then, at **508**, the at least one processing device will determine that a fault is pending with respect to the cooling of the battery **202**. Similarly still, if the thermal load status of a different electrical component (e.g., the CCU **120**, the autonomous driving controller **122**, the power control unit **204**, or the starter-generator **302**) exceeds a threshold for that electrical component (e.g., critical) at **410**, while the coolant system status remains under the critical threshold at **408**, then, at **514**, the at least one processing device will determine that a fault is pending with respect to the cooling of that electrical component. Any of these pending faults can lead to the processing device outputting and/or storing in the memory a third cooling performance indicator specific to that electrical component (e.g., DTC **506**, **512**, or **518**).

(29) In various approaches, for the third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) to be output for each electrical component, the at least one processing device must determine that a second fault pending duration of time exceeds a second fault pending threshold time at **504**, **510**, or **516**. In various embodiments, each electrical component may have a different second fault

pending threshold time. For example, the temperature and thermal load status of the inverter **118** can change rapidly, and its fault pending threshold time may be as short as 15 seconds at **504**. However, the battery **202** may change temperatures much more slowly, and as such, its fault pending threshold time may be 5 minutes or more at **510**. If the second fault pending duration of time exceeds the second fault pending threshold time for that electrical component (at **504**, **510**, or **516**), then the processing device may output the respective third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) in response, at least in part, to also determining that the second fault pending duration of time exceeds the second fault pending threshold time.

(30) In various approaches, the at least one processing device may also be configured to determine that a coolant pump speed exceeds a coolant pump speed threshold. If so, the processing device may output the third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) in response, at least in part, to also determining that the coolant pump speed exceeds the coolant pump speed threshold. This ensures that the cooling issue with the particular electrical component is not caused by a lack of coolant flow due to speed of the coolant pump **112** that is too low. This situation does not represent a problem with the cooling system, but instead represents a problem with the present coolant pump speed setting, which may be remedied by increasing the speed of the coolant pump **112**. Each electrical component may have a different coolant pump speed threshold that is required in order to trigger output of the third cooling performance indicator (e.g., DTC **506**, **512**, or **518**). If the coolant pump speed is below the threshold, then the processing device will block output of the third cooling performance indicator. However, as soon as pump speed increases above the threshold, if the thermal load is still above the threshold for that component (while the coolant system status is still below critical), then the fault pending timing begins, which is then compared against the second fault pending threshold time for that electrical component (at **504**, **510**, or **516**).

(31) FIG. **6** is another logic flow diagram illustrating variations of the functions and methods disclosed in FIG. **5**, which the one or more processing devices of the system may be configured to execute, in accordance with various embodiments. FIG. **6** illustrates additional considerations that may be determined prior to outputting a cooling performance indicator (e.g., DTC **506**, **512**, or **518**) for an individual electrical component. The considerations and determinations illustrated in step **604** of FIG. **6** may occur as part of each step **502**, **508**, and/or **514** prior to determining that a fault is pending (except for steps **618**, **620**, and **622**, which are applicable only to rotating electrical components, such as an electric motor or the starter-generator **302**).

(32) At **602**, the basic conditions for detecting an individual electrical component cooling issue are provided, which are largely duplicative of those shown in FIG. **5**. Specifically, the at least one processing device determines that the coolant system status does not exceed the coolant system status threshold (as in **408** in FIG. **5**), that the thermal load status of at least one electrical component exceeds the thermal load status for that electrical component (as in **410** in FIG. **5**). Further, no cooling component has failed (as in **402** in FIG. which may result instead in the output of the second cooling performance indicator (e.g., DTC **420**), and would block output of one of the third cooling performance indicators (e.g., DTC **506**, **512**, or **518**). If these conditions are true for a particular electrical component, then the analysis moves to consideration of individual conditions for setting such individual component cooling failure indicators.

(33) In various embodiments, a first check is performed at **612**. As part of this first check **612**, at **606**, the at least one processing device determines whether the coolant temperature is above or below a non-critical temperature threshold while the coolant system status does not exceed the coolant system status threshold. Because the coolant system status is also based on the ambient temperature and vehicle speed, a very high coolant temperature may not cause the coolant system status to enter the critical level, for example, if it is very hot outside. Thus, a very high coolant temperature may simply be the result of the environment rather than a failure of the coolant system as a whole, and thus, the coolant system status may remain below critical despite the high coolant temperatures. However, such high coolant temperatures may still be at or near derating

temperatures for various electrical components, and may fail to cool those electrical components so that the thermal load status of that electrical component may exceed the thermal load status threshold (as in **410** in FIG. **5**). However, because this high coolant temperature is caused by the environment rather than a failure with the cooling system, it would be improper to output a cooling performance indicator (e.g., DTC **506**, **512**, or **518**) for that particular component. Thus, in order to output the third cooling performance indicator for a particular electrical component, the coolant temperature must be below a non-critical temperature threshold, which may be different for each electrical component (e.g., based on their derating temperatures). Conversely, if it is determined that the coolant temperature is above the non-critical temperature threshold while the coolant system status does not exceed the coolant system status threshold, then the at least one processing device can prevent output of the third cooling performance indicator.
(34) At **608**, the at least one processing device determines that the coolant pump speed exceeds the coolant pump speed threshold for that particular electrical component, as discussed above.
(35) At **610**, the at least one processing device determines that the electric component experienced a power usage exceeding a high power usage threshold within a period of time directly preceding determining that the thermal load status of the at least one electrical component exceeds a thermal load status threshold. High power usage (e.g., during high acceleration) can generate temporary spikes in temperatures of the electrical components (such that the thermal load status of the electrical component exceeds the threshold for that component). The system accounts for those temporary temperature spikes by allowing the power usage of the component to return to normal and waiting for a cooling period of time to determine if the component is still experiencing a critical thermal load status before outputting the third cooling performance indicator. Put another way, the at least one processing device will responsively prevent output of the third cooling performance indicator for a cooling period of time. Conversely, if the electric component has not experienced a power usage exceeding the high power usage threshold, then the processing device may continue with outputting the third cooling performance indicator (upon meeting the other conditions). The cooling period of time is determined by how long the component takes to reach a state where it has not experienced a high power usage plus the second fault pending threshold time (which may be set individually for each component).
(36) In one approach, the at least one processing device determines the power usage by calculating a mean value of power usage by that electrical component over a period of time (e.g., 30 seconds, though can be different and can be individual for different electrical components). If the mean value of the power usage is greater than a threshold for each individual component, then the processing device increments a power usage counter. If the mean value is below the threshold, then the power usage counter is decremented. If the power usage counter exceeds a counter threshold (e.g., 5), then the processing device determines that the electrical component experienced a power usage exceeding the high power usage threshold. The processing device can also increment or decrement the power usage counter by more than 1 (e.g., 2-5) depending on how high the mean value of power usage was over (or under) the threshold for the timeframe. In various embodiments, power usage of one component (e.g., the inverter **118**) may be inferred from power usage of another component (e.g., the battery **202**).
(37) Assuming these three conditions (**606**, **608**, and **610**, as well as the basic conditions of **602**) are met for a second fault pending time that exceeds the second fault pending threshold time in **624** (which is the same as one of **504**, **510**, or **516**), then the processing device may proceed with outputting the respective third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) at **626**.
(38) In various embodiments, a second check is performed at **616**. As part of this second check **616**, at **614**, the at least one processing device is configured to determine a temperature difference between the coolant temperature and a temperature of the electrical component, and to determine whether the temperature difference exceeds a temperature delta threshold. If there is a high difference between the coolant temperature and the component, then an assumption may be made

that there is a problem with cooling of the electrical component. Conversely, if the difference is low (e.g., because the coolant temperature is very high, even though the coolant system status is not critical), then there the assumption that there is a problem with the cooling system's cooling of that component because the temperatures are close to each other. Accordingly, the at least one processing device may output the third cooling performance indicator in response, at least in part, to also determining that the temperature difference exceeds the temperature delta threshold. Conversely, it may prevent output of the third cooling performance indicator in response, at least in part, to determining that the temperature difference does not exceed the temperature delta threshold.
(39) The considerations for the coolant pump speed in **608** and the experienced electric power usage of the electrical component in **610** are included again in the second check **616**. If these three conditions (**614**, **608**, and **610**, as well as the basic conditions of **602**) are met for a second fault pending time that exceeds the second fault pending threshold time in **624** (which is the same as one of **504**, **510**, or **516**), then the processing device may proceed with outputting the respective third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) at **626**.
(40) A third check **622** may be performed for electrical components that rotate, such as an electric motor or the starter-generator **302**. In various embodiments, a direct temperature reading or thermal load status of such rotating components may not be available. Further, such rotating components may also be cooled by air in addition to the coolant in the circuit **102**, where the air cooling is effected by the rotation of the component. Thus, at **618**, the at least one processing device may determine that a rotational speed of at least one of an electric motor or a starter-generator is above a rotational speed threshold (e.g., such that it is being sufficiently cooled by air). This may be received from the starter-generator, or may be inferred from a known rotational speed of the combustion engine. Also, at **620**, the at least one processing device may determine that a torque of the at least one of the electric motor or the starter-generator is below a torque threshold (e.g., such that it is not generating excessive amounts of heat). If these conditions are met, in addition to the experienced electric power usage being below the threshold in **610** (as well as the basic conditions of **602**), for a second fault pending time that exceeds the second fault pending threshold time in **624** (which is the same as one of **504**, **510**, or **516**), then the processing device may proceed with outputting the respective third cooling performance indicator (e.g., DTC **506**, **512**, or **518**) at **626**.
(41) Additionally, in different approaches, for each different component, the system may be configured to perform only one, two, or all three of the first check **612**, the second check **616**, and/or the third check **622**. Many variations are possible.
(42) So configured, an improved cooling system is disclosed that provides additional details regarding issues within the cooling system, including providing cooling performance indicators that not only indicate a status of the cooling system as a whole, but also provide more specific indications of particular component cooling failures. This leads to greater efficiency in diagnosing and repairing such cooling systems, as well as improved detection of coolant system issues.
(43) The foregoing disclosure has been set forth merely to illustrate the invention and is not intended to be limiting. Since modifications of the disclosed embodiments incorporating the spirit and substance of the invention may occur to persons skilled in the art, the invention should be construed to include everything within the scope of the appended claims and equivalents thereof.

Claims

1. A cooling system for cooling a plurality of electrical components of a vehicle, the cooling system comprising: a cooling circuit comprising a plurality of pathways through which coolant flows; a coolant pump configured to pump the coolant through the cooling circuit; a coolant cooler configured to cool the coolant flowing therethrough; at least one processing device configured to: determine a coolant system status based, at least in part, on a coolant temperature of the coolant in

the cooling system; determine whether the coolant system status exceeds a coolant system status threshold; determine a thermal load status of at least one electrical component of the vehicle cooled by the cooling system; determine whether the thermal load status exceeds a thermal load status threshold; output a first cooling performance indicator in response, at least in part, to determining that the coolant system status exceeds the coolant system status threshold and that the thermal load status of the at least one electrical component exceeds the thermal load status threshold, the first cooling performance indicator indicating an issue with the cooling system; output a third cooling performance indicator in response, at least in part, to determining that the coolant system status does not exceed the coolant system status threshold and that the thermal load status of the at least one electrical component exceeds the thermal load status threshold, the third cooling performance indicator indicating an issue with cooling of the at least one electrical component; and control at least the coolant pump based at least in part on the coolant system status and the thermal load status.

2. The cooling system of claim 1, further comprising: a coolant temperature sensor configured to measure the coolant temperature of the coolant in the cooling system at an exit of the coolant cooler, wherein the at least one processing device is further configured to receive the coolant temperature from the coolant temperature sensor.

3. The cooling system of claim 1, wherein the at least one processing device is further configured to: determine the coolant system status based, at least in part, on the coolant temperature, an ambient temperature, and a vehicle speed.

4. The cooling system of claim 1, further comprising: an electrical component temperature sensor configured to measure a temperature of the at least one electrical component, wherein the at least one processing device is further configured to: receive the temperature of the at least one electrical component; and determine the thermal load status based on the temperature of the at least one electrical component.

5. The cooling system of claim 1, wherein the at least one processing device is further configured to: determine that the coolant system status exceeds the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold for a fault pending duration of time that exceeds a fault pending threshold time; and output the first cooling performance indicator in response, at least in part, to also determining that the fault pending duration of time exceeds the fault pending threshold time.

6. The cooling system of claim 1, wherein the at least one processing device is configured to: output the first cooling performance indicator by setting a diagnostic trouble code in an on-board diagnostics (OBD) system of the vehicle.

7. The cooling system of claim 1, further comprising: a cooling system component of the cooling system, wherein the at least one processing device is further configured to: determine that the cooling system component has failed; output a second cooling performance indicator in response, at least in part, to determining that the cooling system component has failed, the second cooling performance indicator indicating a cooling system component failure; and prevent output of the first cooling performance indicator.

8. The cooling system of claim 7, wherein the cooling system component comprises at least one of a cooling fan, the coolant pump, or a shutter flap.

9. The cooling system of claim 7, wherein the at least one processing device is configured to: determine that the cooling system component has failed for a component failure duration of time that exceeds a component failure pending threshold time; and output the second cooling performance indicator in response, at least in part, to also determining that the component failure duration of time exceeds the component failure threshold time.

10. The cooling system of claim 1, wherein the at least one electrical component of the vehicle cooled by the cooling system comprises at least one of an inverter, a controller, a charge control unit, an autonomous driving control unit, a battery, or a starter-generator.

11. The cooling system of claim 1, wherein the at least one processing device is configured to: determine that the coolant system status does not exceed the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold for a second fault pending duration of time that exceeds a second fault pending threshold time; and output the third cooling performance indicator in response, at least in part, to also determining that the second fault pending duration of time exceeds the second fault pending threshold time.
12. The cooling system of claim 1, further comprising: wherein the at least one processing device is configured to: determine that a coolant pump speed exceeds a coolant pump speed threshold; and output the third cooling performance indicator in response, at least in part, to also determining that the coolant pump speed exceeds the coolant pump speed threshold.
13. The cooling system of claim 1, wherein the at least one processing device is configured to: determine whether the coolant temperature is above a non-critical temperature threshold while the coolant system status does not exceed the coolant system status threshold; and prevent output of the third cooling performance indicator in response, at least in part, to determining that the coolant temperature is above the non-critical temperature threshold while the coolant system status does not exceed the coolant system status threshold.
14. The cooling system of claim 1, wherein the at least one processing device is configured to: determine a temperature difference between the coolant temperature and a temperature of the at least one electrical component; determine whether the temperature difference exceeds a temperature delta threshold; output the third cooling performance indicator in response, at least in part, to also determining that the temperature difference exceeds the temperature delta threshold; and prevent output of the third cooling performance indicator in response, at least in part, to determining that the temperature difference does not exceed the temperature delta threshold.
15. The cooling system of claim 1, wherein the at least one processing device is configured to: determine that the at least one electric component experienced a power usage exceeding a high power usage threshold within a period of time directly preceding determining that the thermal load status of the at least one electrical component exceeds a thermal load status threshold; and responsively prevent output of the third cooling performance indicator for a cooling period of time.
16. The cooling system of claim 1, wherein the at least one processing device is configured to: determine that a rotational speed of at least one of an electric motor or a starter-generator is above a rotational speed threshold; determine that a torque of the at least one of the electric motor or the starter-generator is below a torque threshold; and output the third cooling performance indicator in response, at least in part, to also determining that the rotational speed is above the rotational speed threshold and that the torque is below the torque threshold.
17. A method for outputting a cooling performance indicator, the method comprising: pumping coolant through a cooling circuit comprising a plurality of pathways and a coolant cooler using a coolant pump; determining, by a controller, a coolant system status of a cooling system for cooling a plurality of electrical components of a vehicle based, at least in part, on a coolant temperature of the coolant in the cooling system, an ambient temperature, and a vehicle speed; determining, by the controller, that the coolant system status exceeds a coolant system status threshold; determining, by the controller, a thermal load status of at least one electrical component of the vehicle cooled by the cooling system; determining, by the controller, that the thermal load status exceeds a thermal load status threshold; outputting, by the controller, a cooling performance indicator in response, at least in part, to determining that the coolant system status exceeds the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold, the cooling performance indicator indicating an issue with the cooling system; and controlling at least the coolant pump based at least in part on the coolant system status and the thermal load status.
18. The method of claim 17, further comprising: measuring the coolant temperature of the coolant in the cooling system with a coolant temperature sensor at an exit of the coolant cooler.
19. A method for outputting a cooling performance indicator, the method comprising: determining,

by a controller, a coolant system status of a cooling system for cooling a plurality of electrical components of a vehicle based, at least in part, on a coolant temperature of coolant in the cooling system; determining, by the controller, that the coolant system status does not exceed a coolant system status threshold; determining, by the controller, a thermal load status of at least one electrical component of the vehicle cooled by the cooling system; determining, by the controller, that the thermal load status exceeds a thermal load status threshold; outputting, by the controller, a cooling performance indicator in response, at least in part, to determining that the coolant system status does not exceed the coolant system status threshold and that the thermal load status exceeds the thermal load status threshold, the cooling performance indicator indicating an issue with cooling of the at least one electrical component; and controlling at least a coolant pump of the coolant system based at least in part on the coolant system status and the thermal load status.